

CM0246 Discrete Structures

Representing Graphs and Graph Isomorphism

Andrés Sicard-Ramírez

Universidad EAFIT

Semestre 2014-2

Preliminaries

Textbook

Kenneth H. Rosen ([1988] 2004). Matemática Discreta y sus Aplicaciones.

Convention

The numbers and page numbers assigned to chapters, examples, exercises, figures, quotes, sections and theorems on these slides correspond to the numbers assigned in the textbook.

Graphs

Graphs are **discrete structures** consisting of vertices and edges that connect these vertices.

Graphs

Graphs are **discrete structures** consisting of vertices and edges that connect these vertices.

We need different kinds of graphs for solving problems using graph models.

Graphs

Graphs are **discrete structures** consisting of vertices and edges that connect these vertices.

We need different kinds of graphs for solving problems using graph models.

Example (computer networks)

Graphs

Graphs are **discrete structures** consisting of vertices and edges that connect these vertices.

We need different kinds of graphs for solving problems using graph models.

Example (computer networks)

- A computer network

Graphs

Graphs are **discrete structures** consisting of vertices and edges that connect these vertices.

We need different kinds of graphs for solving problems using graph models.

Example (computer networks)

- A computer network
- A computer network with multiple links between data centers (parallel edges)

Graphs

Graphs are **discrete structures** consisting of vertices and edges that connect these vertices.

We need different kinds of graphs for solving problems using graph models.

Example (computer networks)

- A computer network
- A computer network with multiple links between data centers (parallel edges)
- A computer network with diagnostic links (loops)

Graphs

Graphs are **discrete structures** consisting of vertices and edges that connect these vertices.

We need different kinds of graphs for solving problems using graph models.

Example (computer networks)

- A computer network
- A computer network with multiple links between data centers (parallel edges)
- A computer network with diagnostic links (loops)
- A computer network with one-way links (edges with direction)

Graphs

Graphs are **discrete structures** consisting of vertices and edges that connect these vertices.

We need different kinds of graphs for solving problems using graph models.

Example (computer networks)

- A computer network
- A computer network with multiple links between data centers (parallel edges)
- A computer network with diagnostic links (loops)
- A computer network with one-way links (edges with direction)
- A computer network with multiple one-way links (parallel edges with direction)

Simple Graphs

Remark

In all the definitions related to graphs, the set of vertices is non-empty. Moreover, we shall assume that this set is finite.

Simple Graphs

Remark

In all the definitions related to graphs, the set of vertices is non-empty. Moreover, we shall assume that this set is finite.

Definition

A **simple graph** $G = (V, E)$ consist of V , a set of **vertices**, and E , a set of unordered pairs of distinct elements of V called **edges**.

Simple Graphs

Remark

In all the definitions related to graphs, the set of vertices is non-empty. Moreover, we shall assume that this set is finite.

Definition

A **simple graph** $G = (V, E)$ consist of V , a set of **vertices**, and E , a set of unordered pairs of distinct elements of V called **edges**.

Remark

In a simple graph there is no parallel edges nor loops.

Simple Graphs

Remark

In all the definitions related to graphs, the set of vertices is non-empty. Moreover, we shall assume that this set is finite.

Definition

A **simple graph** $G = (V, E)$ consist of V , a set of **vertices**, and E , a set of unordered pairs of distinct elements of V called **edges**.

Remark

In a simple graph there is no parallel edges nor loops.

Example

Whiteboard.

Multigraphs

Definition

A **multigraph** $G = (V, E)$ consist of set V of vertices, a set E of edges and a function

$$f : E \rightarrow \{ \{u, v\} \mid u, v \in V, u \neq v \}.$$

Multigraphs

Definition

A **multigraph** $G = (V, E)$ consist of set V of vertices, a set E of edges and a function

$$f : E \rightarrow \{ \{u, v\} \mid u, v \in V, u \neq v \}.$$

The edges e_i and e_j are called **parallel edges** if $f(e_i) = f(e_j)$.

Multigraphs

Definition

A **multigraph** $G = (V, E)$ consist of set V of vertices, a set E of edges and a function

$$f : E \rightarrow \{ \{u, v\} \mid u, v \in V, u \neq v \}.$$

The edges e_i and e_j are called **parallel edges** if $f(e_i) = f(e_j)$.

Remark

In a multigraph there is no loops.

Multigraphs

Definition

A **multigraph** $G = (V, E)$ consist of set V of vertices, a set E of edges and a function

$$f : E \rightarrow \{ \{u, v\} \mid u, v \in V, u \neq v \}.$$

The edges e_i and e_j are called **parallel edges** if $f(e_i) = f(e_j)$.

Remark

In a multigraph there is no loops.

Example

Whiteboard.

Directed Graphs

Definition

A **direct graph** $G = (V, E)$ consist of a set of vertices V and a set of edges E that are ordered pairs of elements of V .

Directed Graphs

Definition

A **direct graph** $G = (V, E)$ consist of a set of vertices V and a set of edges E that are ordered pairs of elements of V .

Example

Whiteboard.

Adjacent Vertices and Incident edges

Definition

Two vertices u and v in an undirected graph G are called **adjacent** in G if $\{u, v\}$ is an edge of G . If $e = \{u, v\}$, the edge e is called **incident** with the vertices u and v .

Adjacent Vertices and Incident edges

Definition

Two vertices u and v in an undirected graph G are called **adjacent** in G if $\{u, v\}$ is an edge of G . If $e = \{u, v\}$, the edge e is called **incident** with the vertices u and v .

Example

Whiteboard.

Degrees of the Vertices

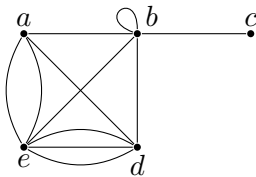
Definition

The **degree of a vertex** v in an undirected graph, denoted $\delta(v)$, is the number of edges incident with it, except that a loop at a vertex contributes twice to the degree of that vertex.

Degrees of the Vertices

Exercise

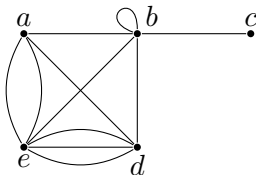
Find the degree of each vertex in the following graph:



Degrees of the Vertices

Exercise

Find the degree of each vertex in the following graph:



Solution

$\delta(a) = 4$, $\delta(b) = \delta(e) = 6$, $\delta(c) = 1$ and $\delta(d) = 5$.

Vertex Degrees

Theorem 1 (the handshaking theorem, p. 511)

Let $G = (V, E)$ be an undirected graph with e edges, then

$$2e = \sum_{v \in V} \delta(v).$$

Vertex Degrees

Theorem 1 (the handshaking theorem, p. 511)

Let $G = (V, E)$ be an undirected graph with e edges, then

$$2e = \sum_{v \in V} \delta(v).$$

Proof.

Each edge contributes **two** to the sum of the degrees of the vertices because an edge is **incident** with exactly two (possibly equal) vertices. This means that the sum of the degrees of the vertices is twice the number of edges. ■

Representing Graphs

- Adjacency matrices
- Incidence matrices

Adjacency Matrices

Definition

Let $G = (V, E)$ be a **simple** graph.

Adjacency Matrices

Definition

Let $G = (V, E)$ be a **simple** graph.

The vertices of G are listed arbitrarily as $v_1, v_2, \dots, v_n$.

Adjacency Matrices

Definition

Let $G = (V, E)$ be a **simple** graph.

The vertices of G are listed arbitrarily as $v_1, v_2, \dots, v_n$.

The **adjacency matrix** $A_G = [a_{ij}]$ of G is a $n \times n$ matrix, where

$$a_{ij} = \begin{cases} 1, & \text{if } \{v_i, v_j\} \text{ is an edge of } G; \\ 0, & \text{otherwise.} \end{cases}$$

Adjacency Matrices

Definition

Let $G = (V, E)$ be a **simple** graph.

The vertices of G are listed arbitrarily as $v_1, v_2, \dots, v_n$.

The **adjacency matrix** $A_G = [a_{ij}]$ of G is a $n \times n$ matrix, where

$$a_{ij} = \begin{cases} 1, & \text{if } \{v_i, v_j\} \text{ is an edge of } G; \\ 0, & \text{otherwise.} \end{cases}$$

Remark

The matrix A_G is Boolean, symmetric and it has zeros in the diagonal.

Adjacency Matrices

Definition

Let $G = (V, E)$ be a **simple** graph.

The vertices of G are listed arbitrarily as $v_1, v_2, \dots, v_n$.

The **adjacency matrix** $A_G = [a_{ij}]$ of G is a $n \times n$ matrix, where

$$a_{ij} = \begin{cases} 1, & \text{if } \{v_i, v_j\} \text{ is an edge of } G; \\ 0, & \text{otherwise.} \end{cases}$$

Remark

The matrix A_G is Boolean, symmetric and it has zeros in the diagonal.

Example

Whiteboard.

Adjacency Matrices

Adjacency matrices for undirected graphs with loops and parallel edges

In this case, the (i, j) -entry represents the number of edges that are associated to $\{v_i, v_j\}$.

Adjacency Matrices

Adjacency matrices for undirected graphs with loops and parallel edges

In this case, the (i, j) -entry represents the number of edges that are associated to $\{v_i, v_j\}$.

Remark

The matrix is symmetric.

Adjacency Matrices

Adjacency matrices for undirected graphs with loops and parallel edges

In this case, the (i, j) -entry represents the number of edges that are associated to $\{v_i, v_j\}$.

Remark

The matrix is symmetric.

Example

Whiteboard.

Adjacency Matrices

Definition

Let $G = (V, E)$ be a **direct** graph where the vertices of G are listed arbitrarily as $v_1, v_2, \dots, v_n$.

Adjacency Matrices

Definition

Let $G = (V, E)$ be a **direct** graph where the vertices of G are listed arbitrarily as $v_1, v_2, \dots, v_n$.

The **adjacency matrix** A_G is defined by

$$a_{ij} = \begin{cases} 1, & \text{if } (v_i, v_j) \text{ is an edge of } G; \\ 0, & \text{otherwise.} \end{cases}$$

Adjacency Matrices

Definition

Let $G = (V, E)$ be a **direct** graph where the vertices of G are listed arbitrarily as $v_1, v_2, \dots, v_n$.

The **adjacency matrix** A_G is defined by

$$a_{ij} = \begin{cases} 1, & \text{if } (v_i, v_j) \text{ is an edge of } G; \\ 0, & \text{otherwise.} \end{cases}$$

Example

Whiteboard.

Incidence Matrices

Definition

Let $G = (V, E)$ be an undirected graph.

Incidence Matrices

Definition

Let $G = (V, E)$ be an undirected graph.

Suppose $v_1, v_2, \dots, v_n$ are the vertices and $e_1, e_2, \dots, e_m$ are the edges.

Incidence Matrices

Definition

Let $G = (V, E)$ be an undirected graph.

Suppose $v_1, v_2, \dots, v_n$ are the vertices and $e_1, e_2, \dots, e_m$ are the edges.

The **incidence matrix** $M_G = [m_{ij}]$ of G is a $n \times m$ **Boolean** matrix, where

$$m_{ij} = \begin{cases} 1, & \text{when } e_j \text{ is incident with } v_i; \\ 0, & \text{otherwise.} \end{cases}$$

Incidence Matrices

Definition

Let $G = (V, E)$ be an undirected graph.

Suppose $v_1, v_2, \dots, v_n$ are the vertices and $e_1, e_2, \dots, e_m$ are the edges.

The **incidence matrix** $M_G = [m_{ij}]$ of G is a $n \times m$ **Boolean** matrix, where

$$m_{ij} = \begin{cases} 1, & \text{when } e_j \text{ is incident with } v_i; \\ 0, & \text{otherwise.} \end{cases}$$

Example

Whiteboard.

Isomorphism of Graphs

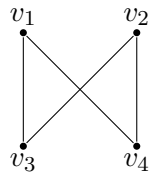
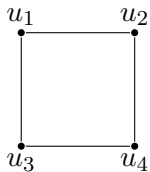
Definition

The simple graphs $G_1 = (V_1, E_1)$ and $G_2 = (V_2, E_2)$ are **isomorphic** if there exists a bijective function f from V_1 to V_2 with the property that u and v are adjacent in G_1 , if and only if, $f(u)$ and $f(v)$ are adjacent in G_2 , for all u and v in V_1 .

Isomorphism of Graphs

Example

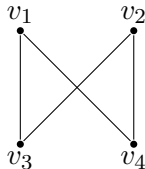
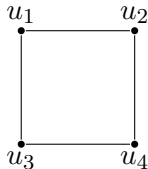
The following simple graphs are isomorphic.



Isomorphism of Graphs

Example

The following simple graphs are isomorphic.



The bijective function f preserves adjacency.

$$f : \{u_1, u_2, u_3, u_4\} \rightarrow \{v_1, v_2, v_3, v_4\}$$

$$f(u_1) = v_1, f(u_2) = v_4, f(u_3) = v_3 \text{ and } f(u_4) = v_2.$$

Isomorphism of Graphs

Remark

Determining whether two simple graphs are isomorphic is often difficult because if $|A| = |B| = n$ then

$$|\{ f : A \rightarrow B \mid f \text{ is a bijection} \}| = n!.$$

Isomorphism of Graphs

Definition

A property preserved by isomorphism of graphs is called a **graph invariant property**.

Isomorphism of Graphs

Definition

A property preserved by isomorphism of graphs is called a **graph invariant property**.

Example

The number of vertices, the number of edges and the degrees of the vertices are graph invariant properties.

Isomorphism of Graphs

Definition

A property preserved by isomorphism of graphs is called a **graph invariant property**.

Example

The number of vertices, the number of edges and the degrees of the vertices are graph invariant properties.

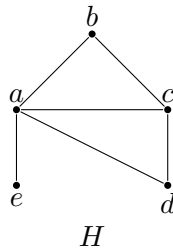
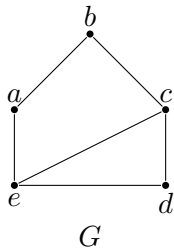
Remark

We can prove that two graphs are not isomorphic if we can find a graph invariant property that only one of the two graphs has.

Isomorphism of Graphs

Example

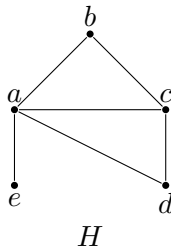
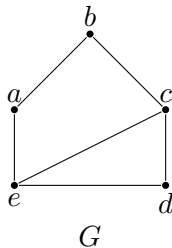
Are the following graphs isomorphic?



Isomorphism of Graphs

Example

Are the following graphs isomorphic?



Solution

No. The graph H has a vertex of degree 1 but the graph G have no vertices of degree 1.

Isomorphism of Graphs

Definition

The **complementary graph** $\overline{G}$ of a simple graph G has the same vertices as G . Two (different) vertices are adjacent in $\overline{G}$ if and only if they are not adjacent in G .

Example

Whiteboard.

Isomorphism of Graphs

Definition

A simple graph G is called **self-complementary** if G and $\overline{G}$ are isomorphic.

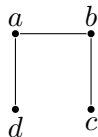
Isomorphism of Graphs

Definition

A simple graph G is called **self-complementary** if G and $\overline{G}$ are isomorphic.

Problem 50 (p. 529)

Is the given graph
self-complementary?



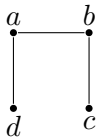
Isomorphism of Graphs

Definition

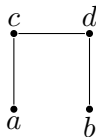
A simple graph G is called **self-complementary** if G and $\overline{G}$ are isomorphic.

Problem 50 (p. 529)

Is the given graph self-complementary?



Yes! The complementary graph is given by the figure.



The isomorphism is $f(a) = c$, $f(b) = d$, $f(c) = b$ and $f(d) = a$.

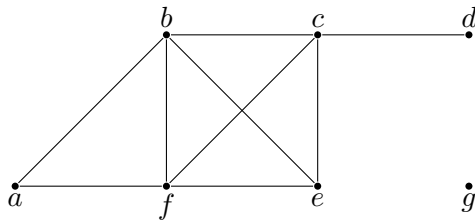
Isomorphism of Graphs

Definition

The **degree sequence** of a graph is the sequence of the degrees of the vertices of the graph in non-increasing order.

Example

For the graph in the figure, the degree sequence is 4, 4, 4, 3, 2, 1, 0.



Isomorphism of Graphs

Problem 69 (p. 530)

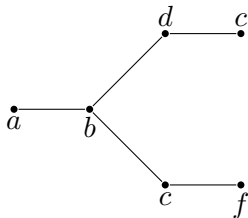
A counter-example for a purported isomorphism test is a pair of non-isomorphic graphs that the test fails to show that they are not isomorphic.

Find a counter-example for the test that checks the degree sequence in two graphs to make sure they agree.

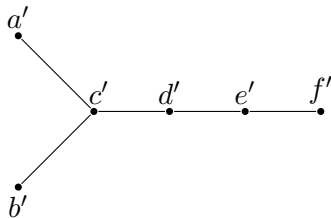
Isomorphism of Graphs

Solution

The degree sequence of both graphs is 3, 2, 2, 1, 1, 1 but they are not isomorphic. In graph G , the vertex b has degree 3 and it is adjacent to two vertices of degree 2 and one vertex of degree 1. The graph H has no vertex with these properties.



G



H

Isomorphism of Graphs

Comparison of several time complexity functions

$f(n)$	10	50	100
$\log n$	2.3 sec	3.9 sec	4.6 sec
n	10 sec	50 sec	1.7 min
n^2	1.7 min	41.7 min	2.8 h
2^n	17.1 min	358.001 c	4×10^{20} c
3^n	16.4 h	2.3×10^{14} c	1.6×10^{38} c
$n!$	42 d	9.7×10^{54} c	3×10^{148} c

Isomorphism of Graphs

Algorithms for graph isomorphism

The best algorithm known has time complexity of $2^{O(\sqrt{n \log n})}$, where n is the number of vertices (Johnson 2005).

Isomorphism of Graphs

Algorithms for graph isomorphism

The best algorithm known has time complexity of $2^{O(\sqrt{n \log n})}$, where n is the number of vertices (Johnson 2005).

vertices	10	100	1000	10000
$2^{\sqrt{n \log n}}$	27.8 sec	33.4 d	3.3×10^{15} c	7.3×10^{81} c

References

- David S. Johnson (2005). The NP-Completeness Column. ACM Transactions on Algorithms 1.1, pp. 160–176. DOI: [10.1145/1077464.1077476](https://doi.org/10.1145/1077464.1077476) (cit. on pp. 61, 62).
- Kenneth H. Rosen [1988] (2004). Matemática Discreta y sus Aplicaciones. Trans. by José Manuel Pérez Morales, Julio Moro Carreño, Ana Isabel Lías Quintero and Pedro Antonio Ramos Alonc. 5th ed. McGraw-Hill (cit. on p. 2).